

Aditya Khajanchi . Jarvis Consulting

I am a curious and enthusiastic Business Systems Analyst with strong communication and collaboration skills. I enjoy taking on new challenges and solving business problems that drive meaningful impact and support organizational goals. Presently at Jarvis, I am applying cloud, big data, analytics, and business systems analysis skills to deliver simulated banking and financial services projects that mirror real-world challenges. Across my academic and professional experiences, I have contributed to cross-functional projects through leadership, stakeholder engagement, and adaptability in dynamic and fast-paced environments.

Skills

Proficient: SQL, PowerBI, Tableau, Excel, PowerQuery, Reporting, Presentations, Git

Competent: Agile/Scrum, Waterfall, Python, Databricks, Hadoop, Spark/PySpark, MS SQL Server, ETL, Looker, Visio, Azure AI Services, GIS Data, Trello, UML Diagrams, Documentation

Familiar: R, Snowflake, JIRA, OpenCV, Firebase, JAVA, MS Project, Technical Writing

Jarvis Projects

Project source code: https://github.com/jarviscanada/jarvis_data_eng_AdityaKhajanchi

PySpark Big Data Analytics: Databricks & Hadoop/Zeppelin Evaluation [GitHub]: Developed a scalable big data analytics solution by re-architecting a single-node Python/Jupyter setup to Apache Spark cluster environments. Implemented retail customer analytics, including Recency-Frequency-Monetary (RFM) segmentation, on Azure Databricks using DBFS, Hive Metastore, and Spark Structured APIs (DataFrames, SQL). Evaluated Spark performance in a Hadoop ecosystem by deploying World Development Indicators (WDI) dataset on GCP Dataproc clusters with HDFS, Hive, and Zeppelin notebooks, analyzing GDP trends across countries. Explored Spark internals (jobs, stages, partitions, Catalyst optimizer) and demonstrated distributed computing advantages by comparing Databricks and Hadoop/Zeppelin implementations. Delivered reproducible notebooks, architecture diagrams, highlighting cloud-based scalability, structured APIs, and practical big data processing workflows.

Hadoop Big Data Analytics Project: Setup and Query Optimization [GitHub]: Implemented a scalable Hadoop cluster on Google Cloud Platform (1 master, 2 worker nodes) to assess Hadoop as an alternative to legacy analytics tools such as SAP and R. Designed and tested distributed storage and processing using HDFS, YARN, and Hive, with interactive analysis in Zeppelin notebooks. Conducted performance benchmarking on a 1TB stock trades dataset, demonstrating how distributed computing reduced execution times from hours to minutes. Optimized Hive queries with year-based partitions, Parquet storage (reducing size from 1.7 GB to 263 MB), and the Tez engine, improving query speeds by up to 45x. Explored MapReduce, Tez, and HiveQL for analytics while comparing imperative and declarative SQL approaches in distributed environments.

Power BI Dashboard Suite: Sales, Survey & Financial Visualizations [GitHub]: Developed three interactive Power BI dashboards in retail, HR, and financial analytics, incorporating advanced data modeling, DAX measures, real-time integrations, and APIs. The projects included a Beverages Sales Dashboard analyzing Coca-Cola product performance with KPIs, geographic insights, and key influencer analysis; a Data Professionals Survey Report visualizing workforce trends from 630 respondents using diverse visual formats; and a Stocks Dashboard leveraging Alpha Vantage API for live market tracking with candlestick charts, key ratios, and dynamic visuals. Each dashboard was designed for decision-makers with a focus on clarity, interactivity, and actionable insights.

Customer Analytics and RFM Segmentation for LGS (PoC) [GitHub]: Designed a system architecture diagram mapping the end-to-end data flow for London Gift Shop (LGS), from raw CSV ingestion to processed insights and dashboard outputs, enabling scalable retail analysis. Developed a Python-based proof of concept for data cleaning, preprocessing, and exploratory analysis using pandas, NumPy, and visualization tools including Matplotlib, Seaborn, Plotly, and Folium. Implemented logic to handle null values, correct data types, and enrich data by mapping countries to continents. Created interactive charts to highlight sales trends, regional patterns, and customer behavior, and built a Recency-Frequency-Monetary (RFM) matrix for customer segmentation and targeted marketing.

BSA Fundamentals [GitHub]: Completed a hands-on Business Systems Analyst training project focused on two core deliverables. First, authored a comprehensive Business Requirements Document (BRD) for a Linux Cluster Monitoring Agent solution to support automated, real-time resource tracking across up to 100 Rocky Linux servers. The BRD defined project scope, functional and non-functional requirements, stakeholder roles, and a cost-benefit analysis to guide development. Second, conducted an in-depth interview with a business stakeholder to gather requirements for a stock

trading application. Based on the insights, developed a System Requirements Document (SRD) for the software development lead, clearly outlining business goals, functional specifications, user needs, and system behavior to support informed development planning.

SQL Project: Database Design & Advanced Querying [GitHub]: Developed a PostgreSQL-based SQL project to strengthen core and advanced SQL skills, covering DDL, DML, DQL, joins, subqueries, aggregates, string manipulation, and window functions. Designed a sample relational database with three linked tables (members, facilities, bookings) in a Dockerized PostgreSQL environment. Completed hands-on exercises such as modifying data, generating insights through complex queries, ranking with window functions, and analyzing monthly booking trends and top facility usage. Applied real-world scenarios like member recommendations and booking pattern analysis while reinforcing query optimization, data transformation, and PostgreSQL function expertise.

Cluster Monitor [GitHub]: Developed an automated monitoring solution for the Linux Cluster Administration (LCA) team to track hardware specifications and real-time resource usage across Rocky Linux servers. Integrated Bash scripts with PostgreSQL to store data persistently, using Docker for containerized deployment and crontab for scheduled execution. Leveraged Git for version control and implemented a scalable system that supports performance analysis, reporting, and future capacity planning. Demonstrated proficiency in Linux system administration, scripting, database management, and DevOps practices.

Highlighted Projects

Interactive Crime Map for Chicago [GitHub]: Developed an interactive Chicago crime heatmap dashboard using Dash and Plotly, allowing users to filter data by crime type, month, or view all available data. Leveraged real-time data from the City of Chicago's public SOD API, utilizing Pandas for data manipulation and Plotly's density_mapbox for visualizing crime density on an interactive map. Implemented dynamic dropdowns, checklists, and callbacks to enable real-time filtering and updates. Integrated Mapbox for precise geographical mapping and customized the user interface with CSS for an enhanced user experience. The dashboard provides a comprehensive view of crime trends and hotspots across the city.

Superstore Dashboard [GitHub]: Collaborated with a team of 7 to develop the Global Superstore Dashboard, an interactive data visualization tool aimed at providing in-depth insights into sales, profit, shipping, and customer segmentation for a global retail operation. Using Dash and Plotly, we built a dynamic dashboard that allows users to explore Year-to-Date (YTD) metrics, such as total sales, orders, and products sold, alongside key performance indicators like sales growth and shipping time. Leveraging Pandas for data processing, we incorporated advanced charts, including area charts, sunburst, funnel, and waterfall plots, to visually present trends, geographical distributions, and customer behavior. The dashboard also integrates dynamic filters, allowing users to select different report types and years for flexible data exploration. This project successfully delivered an intuitive platform for management to analyze real-time business performance and make informed decisions.

Professional Experiences

Business Systems Analyst, Jarvis (2025-present): Participated in the Business Systems Analyst stream of Jarvis's Talent Incubation (TIC) program, simulating real-world business environments. Developed key competencies in business analysis, including requirements gathering, documentation, and stakeholder engagement. Acquired technical skills in relational database management, cloud computing, Linux scripting, Python programming, PowerBI and Hadoop Distributed File Systems. Applied project management methodologies such as Agile, Scrum, Waterfall, and Kanban to manage deliverables in team-based scenarios. Collaborated with cross-functional peers to solve business problems and deliver structured documentation under simulated workplace timelines. Strengthened communication and critical thinking skills through iterative feedback and peer reviews.

Data Analyst, Geopogo (2022-2025): Collaborated to develop a Tableau dashboard displaying KPIs from 12+ data sources for stakeholders, including investors, senior management, and team leads. Implemented an ETL pipeline to automate data transformation, reducing manual data entry by 50% and saving up to 5 hours weekly. Streamlined data management for ZoneQuestAI by implementing batch imports/exports and scheduled backups in Firebase, enhancing process efficiency, data integrity, and accessibility. Identified key usage and engagement trends that led to a 15% increase in engagement and 25% increase in organic growth through data-driven recommendations. Contributed to the research, extraction, and implementation of GIS data from publicly available sources to integrate into the latest release.

Analyst, Coordinator, Software Engineering, Architectura Architects (2021-2022): Implemented an Excel productivity tracker for LiDAR scanning projects, reducing project turnaround time by 35%. Developed and maintained

a Power BI dashboard, presenting insights on operational efficiency and project timelines to senior management in weekly meetings. Coordinated 25+ projects, managing initiatives over \$10MM for The City of Windsor and \$30MM for Ontario School Boards. Liaised with internal and external project stakeholders to improve process efficiency. Maintained clear and concise documentation and user guides for knowledge transfer.

Education

Zekelman School of Business, St. Clair College (2019-2020), Post-graduate Certificate, Data Analytics - GPA: 3.9/4.0 - Presented 'Rent Analysis & Municipal Services Call Analysis of Windsor City' Data Visualization project to ambassadors of Argentina, Chile, Ecuador, and Peru - Class Representative, Student Council

School of Computer Studies, Ahmedabad University (2015-2018), Bachelor of Computer Application, Computer Applications - GPA: 3.8/4.33 - President, Computer and Tech Club - President, Campus Events Committee - Class Representative, Student Council

Miscellaneous

- IBM, Certified Data Analyst (2024)
- Microsoft, Azure AI Fundamentals [AI-900] (2024)
- Microsoft, Certified Power BI Data Analyst Associate [PL300] (Expected 2025)
- Google, Foundations of Project Management (Expected 2025)
- Winner, Best Data Visualization Project, St. Clair College (2019)
- I enjoy participating in professional seminars and networking events to stay updated in my field and connect with like-minded professionals, which helps me stay current with industry trends and continuously refine my communication and interpersonal skills.
- I like tackling DIY projects every other weekend, whether it's around the house, working on my car, or tech-related fixes, as they sharpen my troubleshooting, creativity, and problem-solving abilities, skills I often carry into my professional work.
- I love experimenting with fusion cooking for my friends and family, creating unique dishes that blend different cuisines and flavors. It has strengthened my patience, attention to detail, and flexibility while also teaching me how to improvise and create something great out of minimal resources.